Advanced Business Analytics
K62 Digital business

GROUP 1

VIETNAM REAL ESTATE AVM

An independent estimate of fair value for every resale home in Hanoi and HCMC

60%–64%

Apartment and alley-house estimates within ±20% of asking price

1.5×

Top 10% of risk scores:
stale rate vs the test-set average
(unsold for over 180 days)

VND 9.7 bn

Net partner value per year
Base-case estimate

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01	**Problem & market** Why a shared pricing reference is needed	03–04
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THE PROBLEM

Homes cost billions of VND, but talks start from the seller's asking price.

Brokers compare similar homes by hand. Banks wait for manual valuations.

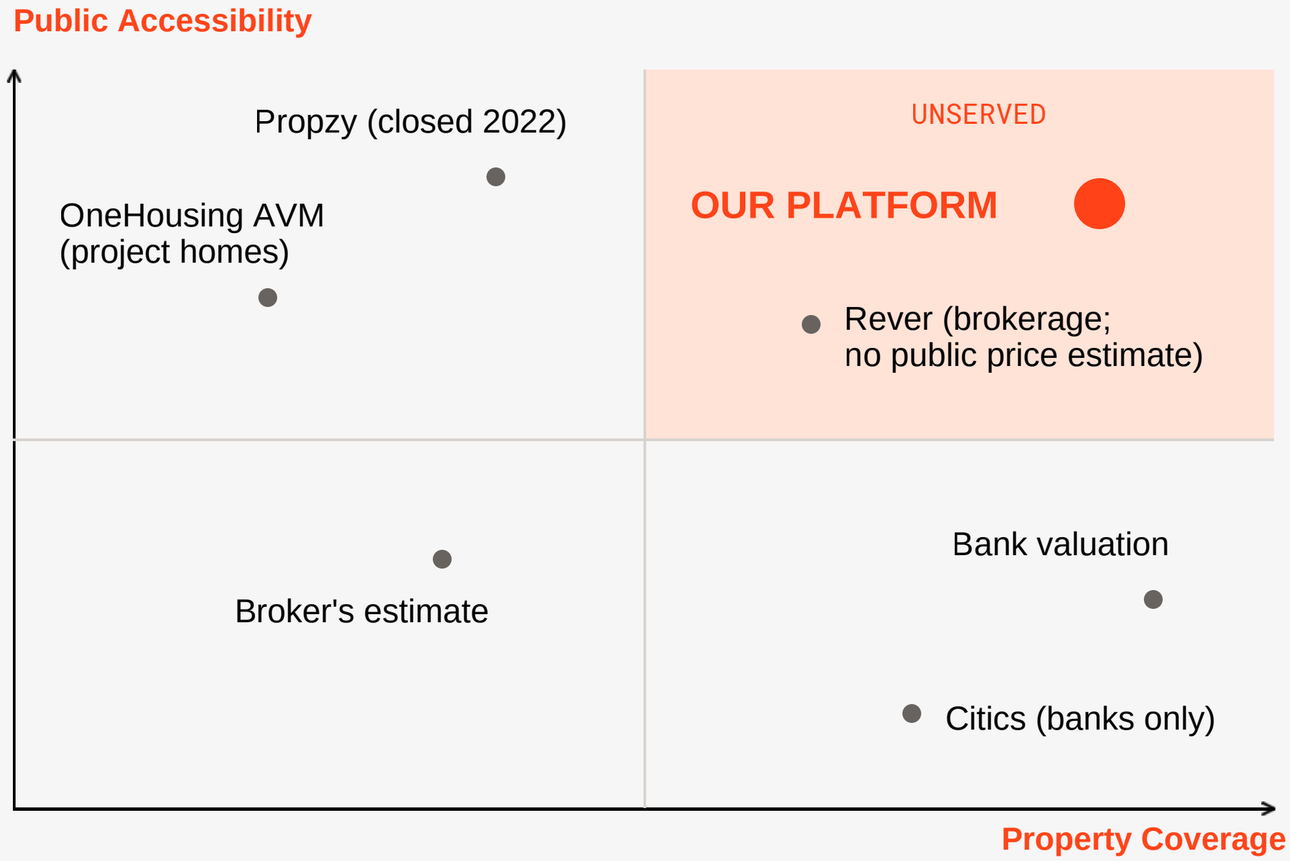
The missing price guide hurts four teams in different ways.



Source: team analysis.

Public values for all property types.

MARKET POSITIONING



OneHousing

533,856 homes in 627 projects one month after the 2022 launch. Reports error below 5%. Techcombank partner.

Citics

More than 32 banks use it. Raised USD 2.1 million in 2025. For institutions only.

Rever is a broker, with no public valuation. Propzy closed in 2022.

Public values for all property types. Method and error by type are published. Uses listings and map data.

Source: company websites and media reports; claims are not independently verified.

One table supports three decisions.

THE SYSTEM

P

Problem

Buyers, brokers and lenders in Hanoi and HCMC have no independent price guide for resale homes.

D

Data

5,268 mogi.vn sale listings linked to 34,321 OpenStreetMap points: **metro, schools, parks, markets and city centre.**

M

Method

Fair Value Engine
Regression + gradient boosting
Stale Listing Radar
Unsold-risk model
Market Map
District groups

T

Tool

Reproducible local workflow and browser-based dashboards.
Planned: cloud database, scheduling, monitoring, and model approval.

DECISIONS

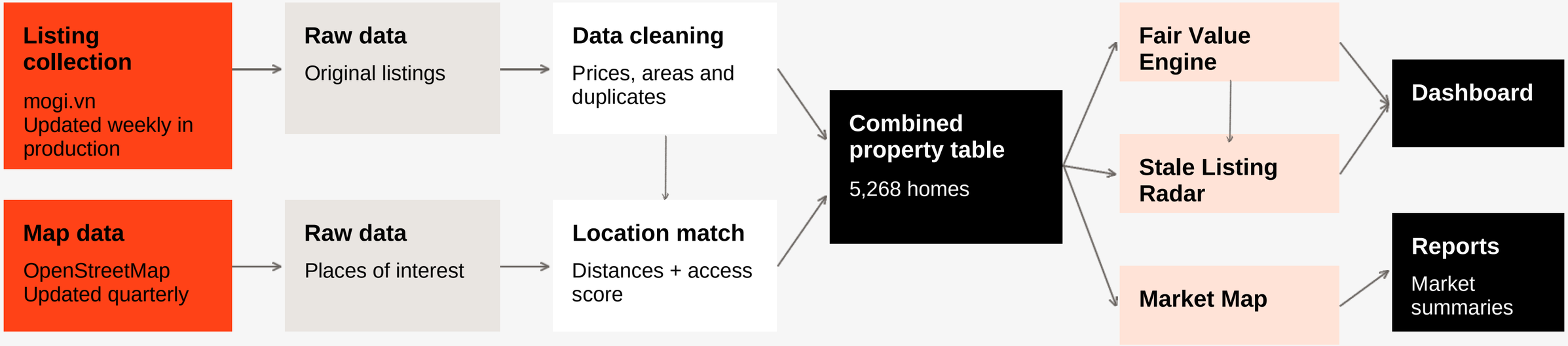
Fair value
+ 90% range
Unsold-risk rank
District group

Price, advise sellers, lend and allocate

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268).

5,268 homes linked to 34,321 map points.

DATA PIPELINE



6,000 listings in 46 minutes. 88% pass data checks.

Prototype collection window: 1.5 hours. Speed: 7,792 listings/hour, paced request. About 33,000 listings take about 5 hours.

Prototype: laptop files. Production: cloud database with location queries, scheduled runs and model versions with accuracy records.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); OpenStreetMap contributors (ODbL).

Three linked engines support three decisions.

MODEL ARCHITECTURE

Combined property data

Size, rooms, legal status, distances and district

Fair Value Engine

Regression + boosted trees; a 90% price range

→ **Set prices; screen loans**

Fair value + range; first collateral check

↓

Stale Listing Radar

Predicts risk of staying unsold over 180 days

→ **Prioritise; advise sellers**

Weekly list of high-risk listings

↓

Market Map

Groups districts by their market

→ **Target areas; assign brokers**

Where to advertise and add brokers

All accuracy scores use homes unseen in training. Age starts when a listing is created: sellers change posted dates, and 59% re-posted more than 30 days after the photos were uploaded

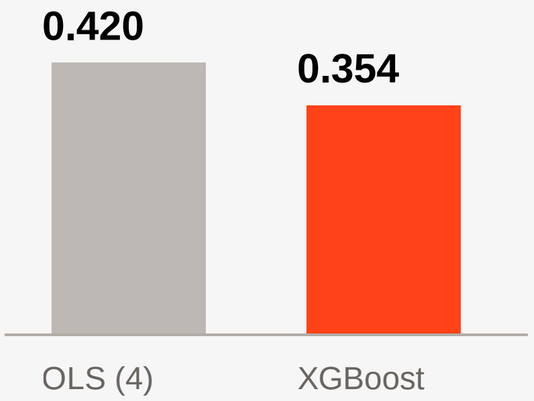
53.9% of 1,054 unseen homes are valued within ±20% of asking price.

FAIR VALUE ENGINE

XGBoost beats regression on all four measures

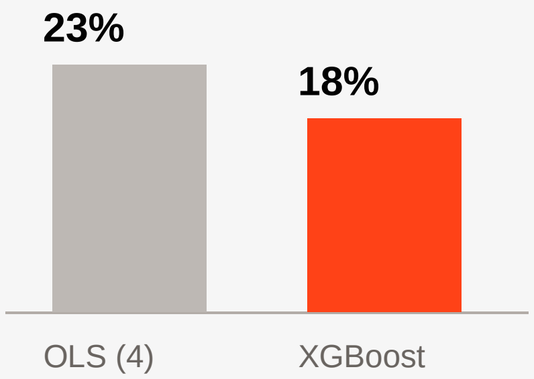
Log-price error (RMSE)

LOWER IS BETTER



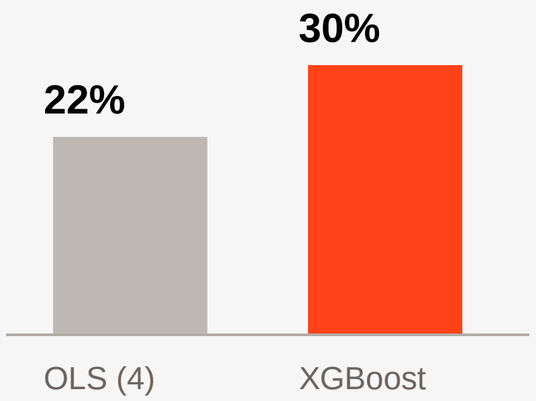
Median absolute % error

LOWER IS BETTER



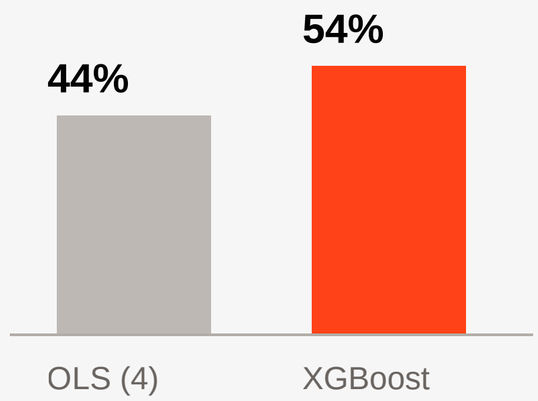
Within ±10% (PPE10)

HIGHER IS BETTER



Within ±20% (PPE20)

HIGHER IS BETTER



Ready for broker use

Within ±20%: alley houses 64%, apartments 60%. Street-front houses 42% and land 43% need more work.

Always show the 90% price range

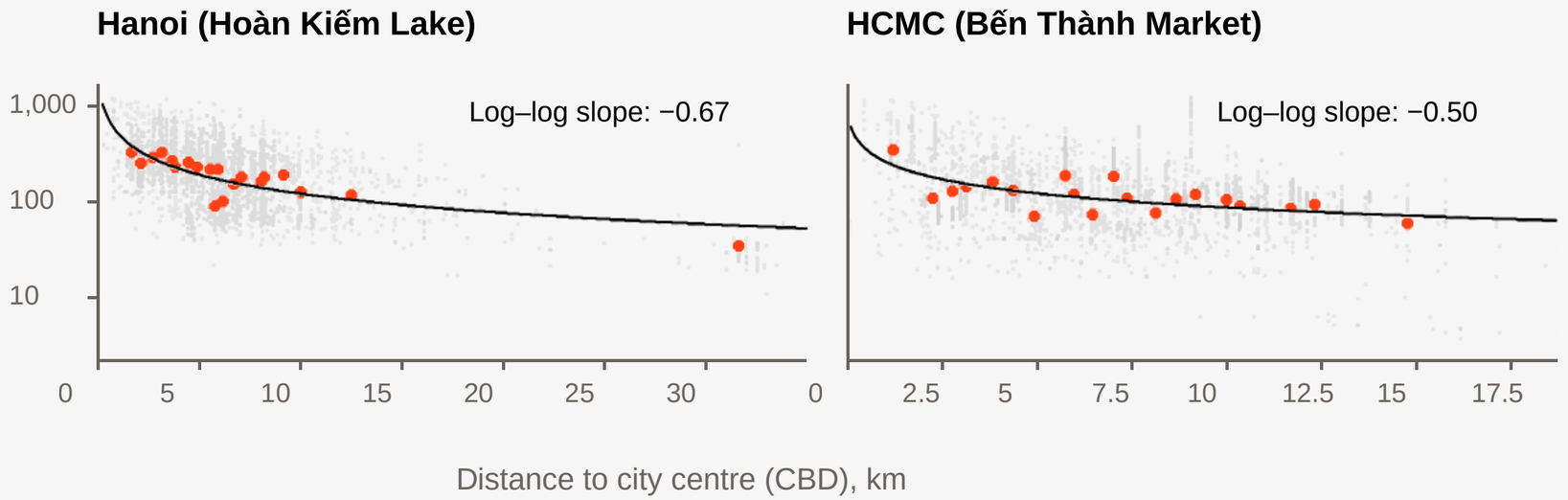
Half of estimates miss by over 18.1%. Log-price error is 0.354; the target is 0.20.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); 1,054 test homes unseen in training. PPE20 bar rounded to 54%.

Map data gives 60% of the accuracy gain from district data.

MAP DATA

Asking price (million VND/m²; log scale)



Points and trend lines redrawn from the source chart

R² 0.714

Fit: size, rooms and legal status

Adding district cuts error by 25%.

Map distances alone cut error by 15%.

Values homes in districts with few listings.

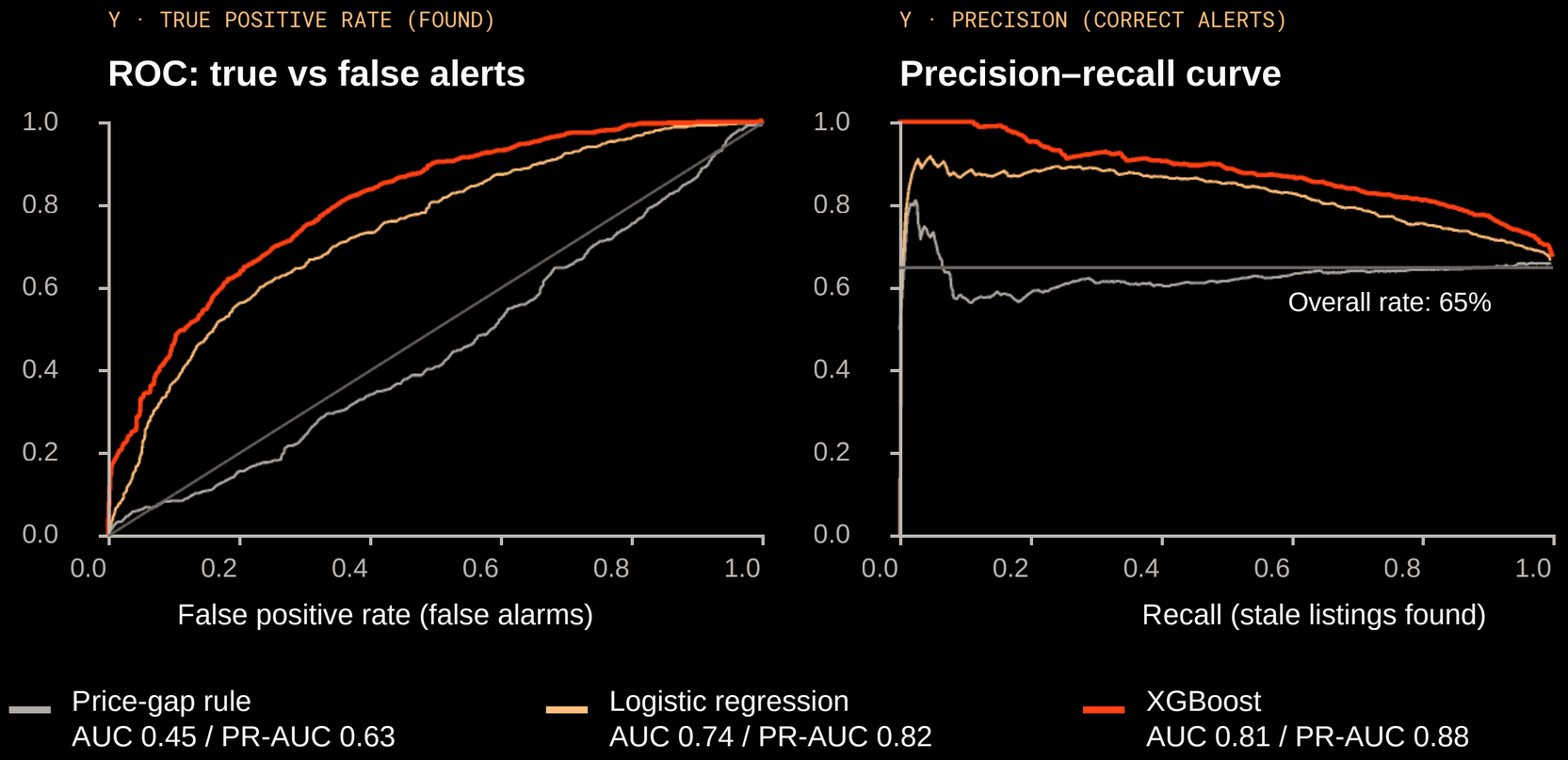
Map data helps value homes where listings are scarce

Source: mogi.vn, 15 Oct 2026 (N = 5,268); OpenStreetMap (ODbL). Regression uses log asking prices; error checked on held-out folds (cross-validation).

The top 10% have 1.5× the average stale rate: unsold for over 180 days.

STALE LISTING RADAR

Top 10%: 99% unsold over 180 days; 65% across all test listings.



AUC 0.81

AUC = ranking score; 0.5 = guessing

Price-gap rule: no better than guessing. Price gap ranks 12 of 26 risk factors.

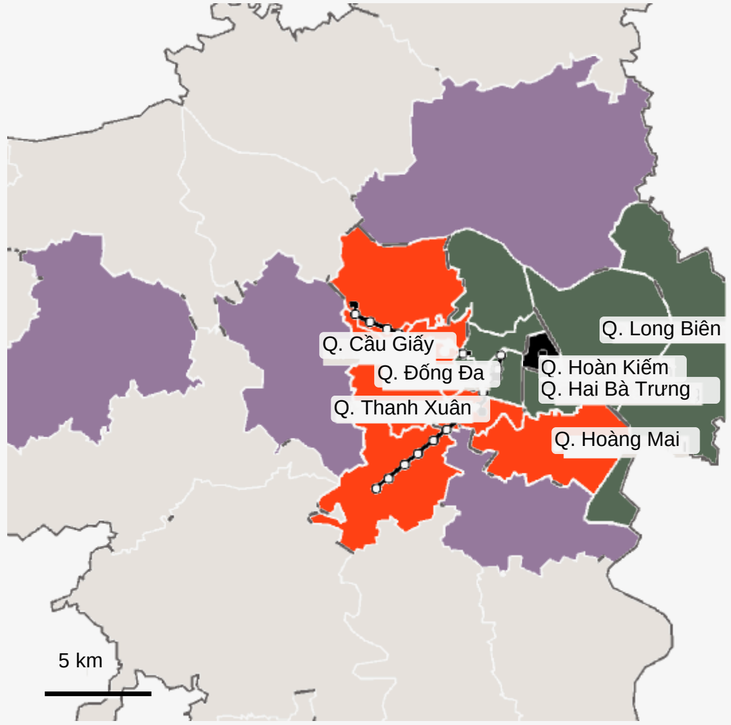
Use radar to focus work. Use price gap to advise sellers.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); 1,054 test listings. Curves redrawn from the published chart.

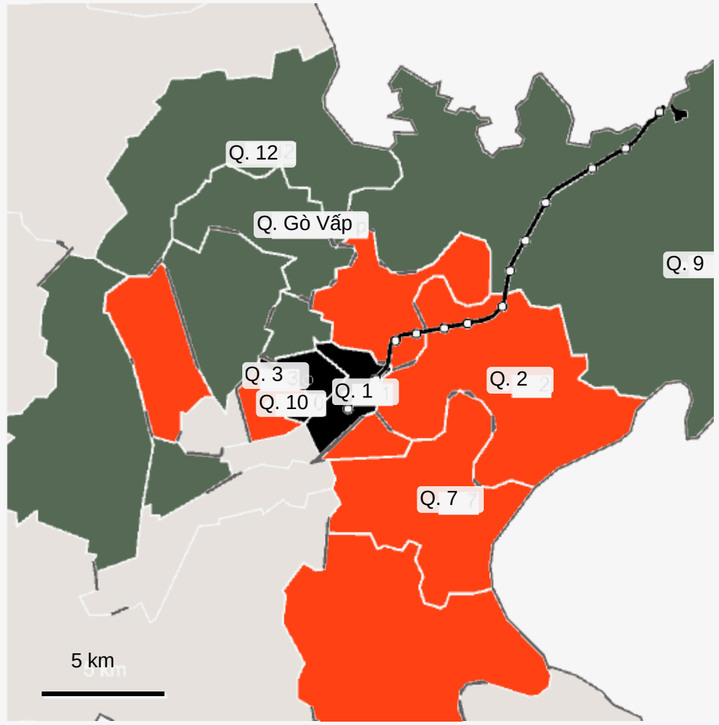
Four district groups need four different messages.

MARKET MAP

Hanoi



Ho Chi Minh City



Prime core
Peripheral growth
Apartment mid-market
Too few listings (< 30)
Townhouse mid-market
Active metro line and station

01 Prime core

415 million VND/m². Hold price.

02 Apartment mid-market

Slowest: 364 days. Increase sales effort.

03 Townhouse mid-market

Fastest: 254 days. Sell more listings.

04 Peripheral growth

Mostly land. Highest stale risk: 36%.

Groups overlap. Use them for targeting, not fixed borders.

Source: mogi.vn, 15 Oct 2026 (N = 5,268); OpenStreetMap contributors (ODbL); district boundaries GADM 4.1.

TOOLS

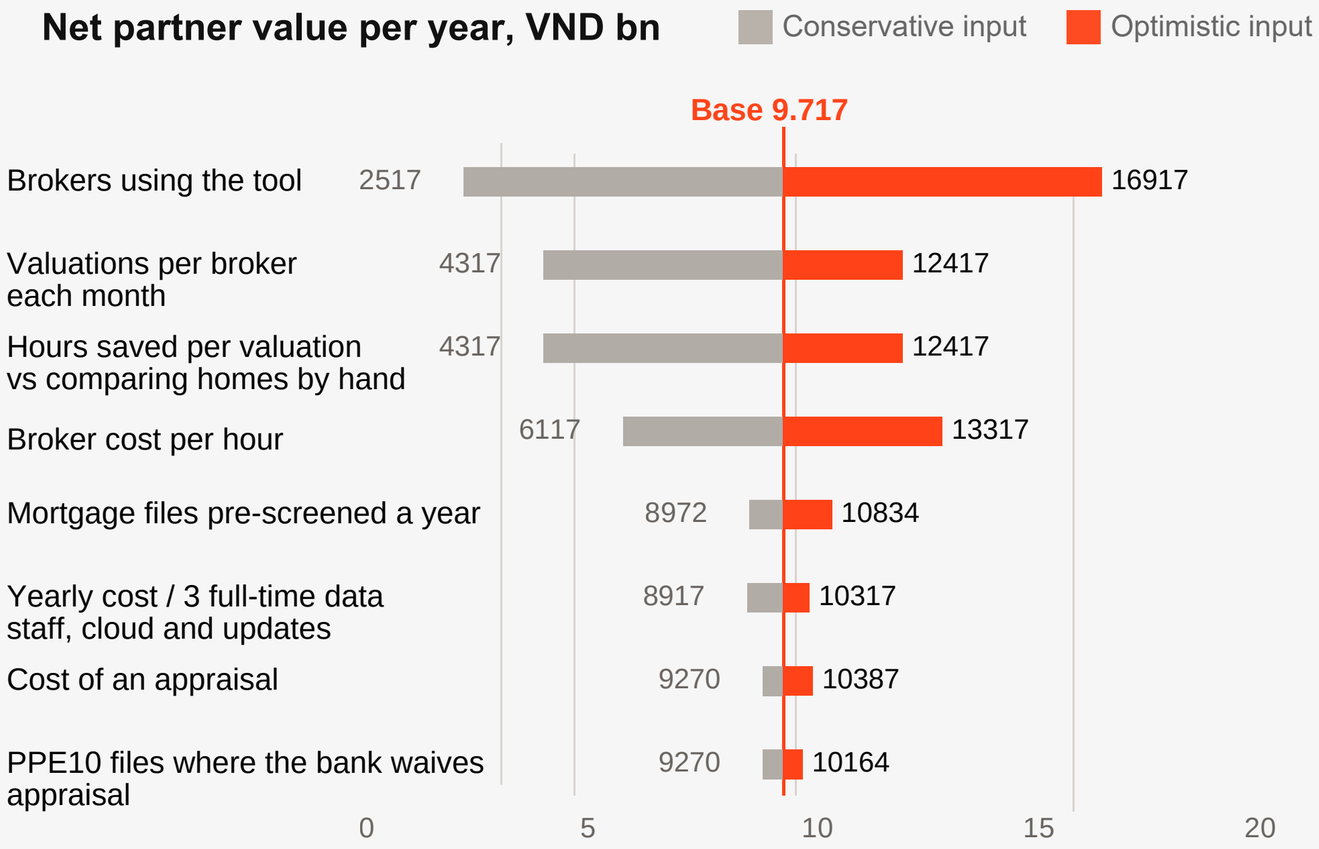
SELECTION CRITERIA

- **Scalability**
- **Collaboration & Reproducibility**
- **Manager Usability**
- **Cost Efficiency**
- **Explainability**
- **Data Privacy & Access**

LAYER	TOOLS	OUTPUTS	WHY IT FITS
01 **DATA COLLECTION**	Python (requests, BeautifulSoup) OpenStreetMap / Overpass · GADM	Property listings, amenities and administrative boundaries	**Scale & cost:** 7,792 listings/hour; lower paid map API costs.
02 **STORAGE**	CSV · JSON · Parquet Planned: cloud spatial database	Integrated dataset of 5,268 homes	**Scalability:** local prototype; planned cloud for shared access.
03 **PROCESSING & MODELING**	pandas · GeoPandas · scikit-learn statsmodels · XGBoost · SHAP	Valuations + 90% intervals Stale-listing risk · Segments	**Explainability:** regression baseline; SHAP explains XGBoost estimates.
04 **PRESENTATION**	Streamlit · HTML/JavaScript matplotlib · Folium	Valuation dashboard Market and district summaries	**Usability:** no-code access; HTML matched Python on 500 listings.
05 **OPERATIONS & GOVERNANCE**	run_all.py · pytest Planned: scheduling & monitoring	Reproducible pipeline Planned model oversight	**Reproducibility:** automated tests; planned model-change approvals.

Adoption drives value: VND 9.7 bn net a year in the base case, 4.4× run cost.

VALUE AND ROI



9.7

VND million net partner value a year

= 4.4× the run cost

Partner value 11.9 (broker time 10.8 + bank appraisal 1.1) − run cost 2.2

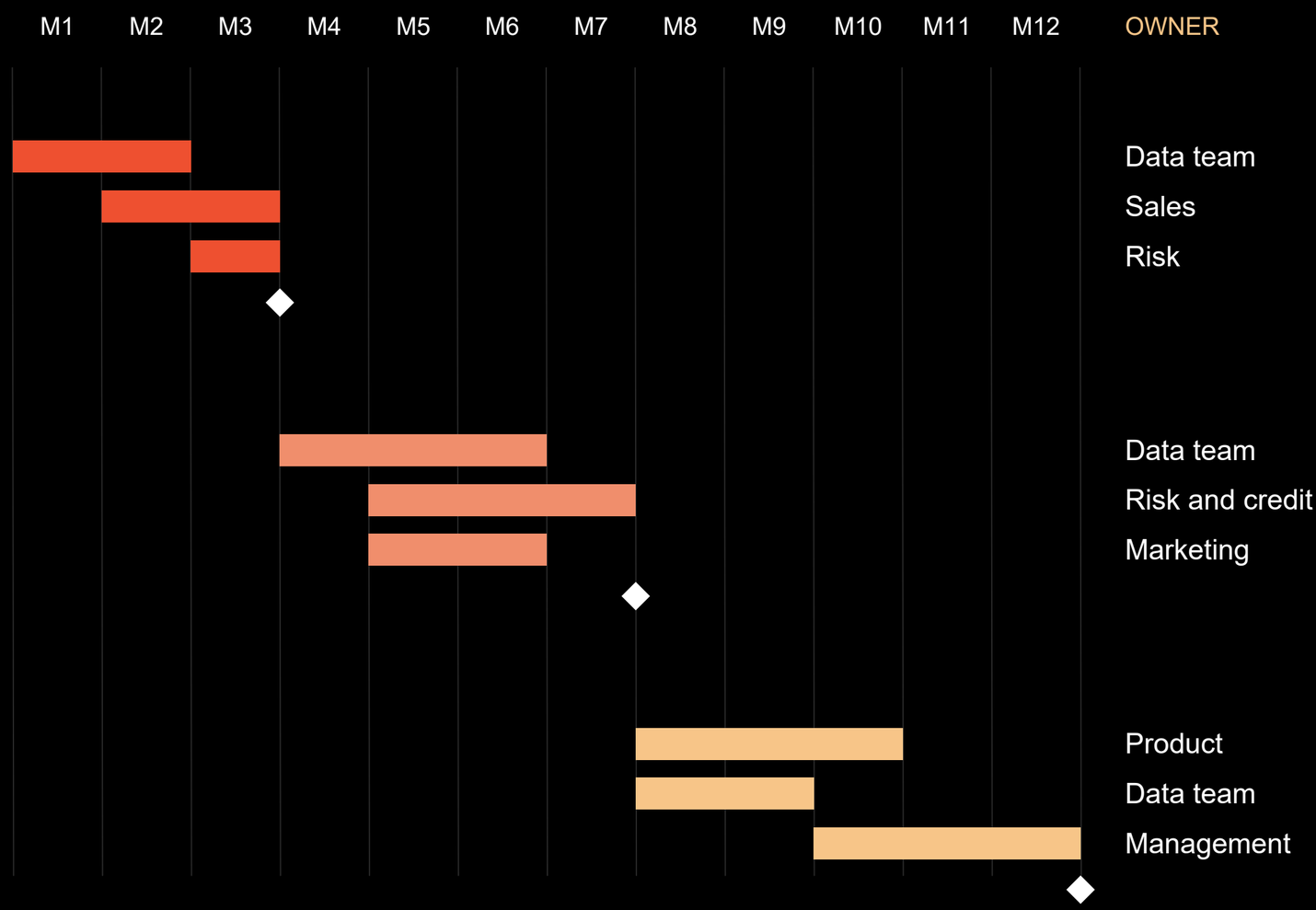
- Range, all inputs conservative → all optimistic: VND −2.3 to 40.9 bn
- Break-even at base inputs: about 30 active brokers with the bank saving, about 60 without
- Broker time is 91% of partner value
- Buyers avoid about VND 10.4 bn of overpayment (scenario, outside the ROI)
- Steady state at 300 active brokers; year one is lower

Remember: broker adoption is the largest driver of projected value. The base case assumes 300 active brokers; break-even is about 60 without bank savings.

Scenario estimates on stated assumptions, not measured results. The 4.4× is an estimated benefit-to-cost ratio: broker time savings are monetised productivity, not necessarily cash. Valuations per broker and hours saved have the same effect because both multiply broker time saved.

12 months, three phases, and a continue-or-stop decision at each step.

IMPLEMENTATION ROADMAP



01 PILOT / M1–M3

Weekly data collection and database	Data team
Broker pilot in one agency	Sales
Monthly accuracy report	Risk

Gate 1 (M3): 60%+ within ±20% on pilot deals; weekly broker use

02 SCALE / M4–M7

Second portal and transaction prices	Data team
Bank pre-screening trial	Risk and credit
Segment reports for campaigns	Marketing

Gate 2 (M7): Bank waiver rule agreed; 300 active brokers

03 PRODUCTISE / M8–M12

Public valuation page and API	Product
Model version tracking, drift alerts	Data team
Quarterly market index for executives	Management

Gate 3 (M12): Accuracy published, stable 3 months; legal review done

Scale only when the numbers say so: accuracy and usage, then the bank agreement and 300 brokers, then published accuracy and legal clearance.

Six known risks and three requests for a 3-month pilot.

RISKS AND PILOT ASK

MAIN RISKS + RESPONSES

Asking prices vs sale prices — H / H
Calibrate on bank appraisals in Phase 2.

Brokers adopt too slowly — M / H
Track active brokers monthly; Gate 2 needs 300.

Noisy unsold-listing label — H / M
Weekly collection gives true time to sale.

Website changes or blocks — M / H
Tested page readers; second portal.

Accuracy drifts as prices move — M / M
Monthly report; re-train if PPE20 drops 5 points.

Publishing home values — L / H
Ranges, disclaimers, published method.

TO START THE PILOT

01 One brokerage for a 3-month pilot (apartments, alley houses). Track active brokers, valuations per broker and hours saved.

02 Appraised or notarised sale prices from a partner bank.

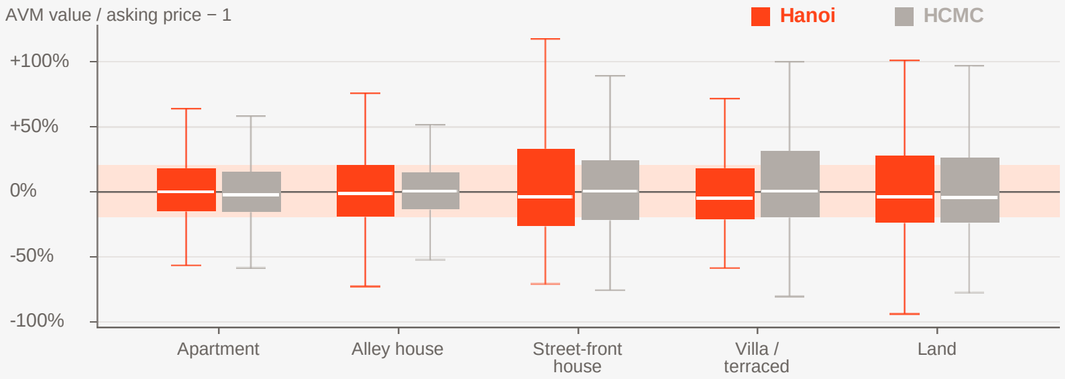
03 Fund a 3-person data team: VND 2.2 bn a year in the base case.

3-MONTH PILOT

APPENDIX

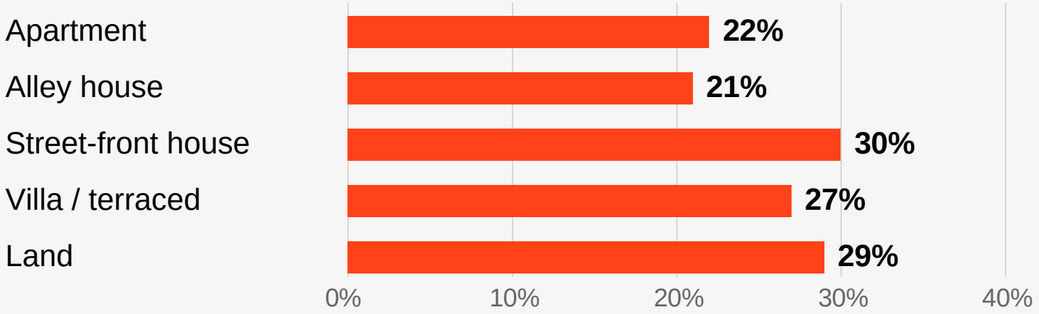
Expensive homes tend to be undervalued. Check this before any lending use.

Errors on homes not used for training



Box: middle 50% (P25–P75). Shading: ±20%.

Loan-security overvaluation risk



Share of homes with AVM value >20% above asking price

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); dashboard prototype.

Vietnam Real Estate AVM

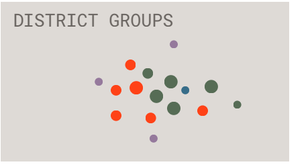
Prototype trained on mogi.vn asking prices (Oct 2026) and OpenStreetMap data. A guide, not a formal appraisal.

Estimated fair value	90% value range	Value per m²
10.89 bn	**6.85–17.34**	**210 m**
VND	bn VND	VND / m²

Location data uses the ward median (middle value) of listed homes.

District group

Townhouse mid-market: median 269 million VND/m², median age 259 days, 238 listings.



Check the asking price

Asking vs fair value	Stale risk (>180 days)
−12.8%	**Low (75%)**

Ranks listing age; not the chance of a sale.

Property

City	Hà Nội
District	Quận Đống Đa
Ward	Select a ward
Type	Alley house
Area	52.00 m²
Bedrooms	4

Similar listings

District	Ward	m²	Beds	VND bn	m VND/m²	Deed	Listed
Quận Đống Đa	Xã Đàn	50.6	4	15.5	306.3241	Sổ đỏ	15/10/2026
Quận Đống Đa	Phường Văn Chương	52	4	21	403.8462	Sổ đỏ	28/05/2026
Quận Đống Đa	Phường Khâm Thiên	48	4	5.5	114.5833	Sổ đỏ	17/10/2024
Quận Đống Đa	Xã Đàn	45	4	7.35	163.3333	Sổ đỏ	16/10/2024
Quận Đống Đa	Phường Ô Chợ Dừa	49	3	9.3	189.7959	Sổ đỏ	28/05/2026

Value, range, price gap, unsold risk, similar homes.

CONTRIBUTIONS

FULL NAME	STUDENT ID	KEY RESPONSIBILITIES	CONTRIBUTION
Đinh Thị Ngọc Anh	2313560003	Stale-listing model; Tools section; Presentation	**12.5%**
Vũ Ngọc Ánh	2313560005	EDA & cleaning; Stale-listing model; Communication & Strategy; Presentation	**12.5%**
Đào Khánh Chi	2313560008	Data section; EDA & cleaning; Fair-value model (XGBoost); Report editing & formatting	**12.5%**
Nguyễn Thảo Linh	2312560022	Fair-value model (OLS); HTML dashboard; Communication & Strategy; Presentation	**12.5%**
Vũ Hoàng Phong	2312560034	Problem statement; Stale-listing model; Prototype & slide editing; Conclusion	**12.5%**
Nguyễn Hà Phương	2313560036	Problem statement; Executive summary & report format; Fair-value model (XGBoost); HTML dashboard	**12.5%**
Trần Minh Huyền	2213560025	Data crawling; Method section; Market Map (K-Means)	**12.5%**
Hà Lê Hoàng	2212560020	Data crawling; Method section; Market Map; Final report check	**12.5%**

TOTAL TEAM CONTRIBUTION **100%**

GROUP 1

THANK YOU